Specification Sheet 3

Job 03: Develop a Location Sharing Application

Time: 2 hours

UNITS OF COMPETENCY (UOC) COVERED

- **OU-ICT-AMAD-05-L4-V1:** Perform Application Programming Interface integration
- **OU-ICT-AMAD-06-L4-V1:** Implement Background Service and Application Deployment
- **OU-ICT-AMAD-07-L4-V1:** Perform Project Work with Android

SPECIFIC INSTRUCTIONS & APPLICATION FEATURES

Specific Instructions:
Build an app that allows users to share their real-time location with friends. Firebase Authentication ensures user privacy, Realtime Database stores location data. Show all locations on Google Maps using markers.

Application Description/Features:
Application will start with a Login/Registration screen. After authentication, the device's current or last location should be detected and user information should be saved on the Database. After that, the user should land on a page where all the users will be shown in a RecyclerView, each item will show the user's name and email address. There should be a page where Google Map will be shown with all the users location as a marker. All the markers should show a snippet where the user name should be displayed. There should be a Profile Screen where the user information will be shown. User should also be able to update his display name and location.

Project Requirements:
Use proper Architectural Components or Android Jetpack Components. For example, View Model, Live Data, Data Binding etc. Use MVVM.

JOB SPECIFICATION INFORMATION

1. Create a new Android Studio project and name it as **LocationSharingApp**.
2. Create a new Firebase Project and connect Android app to Firebase.
3. Enable Authentication (Email and Password) and Cloud Firestore in your Firebase Project.
4. Change Cloud Firestore Database rules, so that no one can access databases without authentication.
5. Add necessary dependencies — Firebase Authentication, Cloud Firestore, Google Play Services for Map and Location.
6. Add necessary permissions for location detection in Manifest file.
7. Get Map API key from cloud console and add the key as meta data in Manifest file.
8. Create an **AuthActivity**, this will be your launcher Activity.
9. Create a **FriendListActivity**. Later, you have to show all your friends in a RecyclerView.
10. Create a **MyProfileActivity**. Here user (who is currently logged in) information (display name, email, latitude, longitude) will be shown.
11. Create a **GoogleMapActivity**. Here you will show all users with markers later.
12. Login or Register (if using this app for the first time) with Email address and Password.
13. After successful login/register, the location detection process will start.

14. Get the device's current location/last location using the `FusedLocationProviderClient` API and retrieve the Latitude and Longitude.
15. Create a Kotlin Data Class, name it as `AppUser` and in the primary constructor, declare the following parameters — `userId`, `userEmail`, `displayName` (nullable), `latitude`, `longitude`.
16. Create an `AppUser` object by providing all the information in the constructor.
17. Save `AppUser` to cloud firestore under `AppUsers` collection.
18. Go to **FriendListActivity** after saving the user.
19. In **FriendListActivity**, show all users in a RecyclerView. Each RecyclerView item should show the User Name and Email Address. If User Name is Null, show a default Text.
20. Create a Floating Options Menu, add three menu items — **My Profile**, **Show users on Map**, and **Logout**.
21. Clicking **My Profile** menu item should take the user to My Profile Page. Show the logged in user information in My Profile page.
22. Provide a button to update the display name (which will be initially null) in My Profile page.
23. Clicking **Show Users on Map** menu item should take the user to Google Map Screen where all the users will be shown on map as Marker. By clicking the marker, show the user name.
24. Clicking the **Logout** menu item should take the user back to the Login Screen.

WORKING PROCEDURE / STEPS

1. Follow OSH and use Personal Protective Equipment (PPE).
2. Check Electricity & Internet Connections to your Computer.
3. Start the Computer.
4. Name a folder on the desktop with your registration number and full name (`ID_FullName`).
5. Read and follow the Specification Sheet.
6. Collect the resources from your assessor as per the job requirement.
7. Open Android Studio, Emulator or real Android device.
8. Create a new Android Studio project.
9. Run app on Emulator/Real device.
10. Zip the project for submission.
11. Close Emulator, Android Studio.
12. Turn off the computer and clean your workplace.

RESOURCE & EQUIPMENT REQUIREMENTS

1. List of Required PPE

S/N	Name of PPE	Specification	Unit	Qty
01	Ergonomic Chair	Wood and foam	No	1
02	Eye protective glass	Metal and Glass	No	1
03	Rubber shoe	Rubber	Pair	1

2. List of Required Materials

S/N	Name of Materials	Specification	Unit	Qty
01	Android Emulator/Real Device	API 21 and higher	—	1

S/N	Name of Materials	Specification	Unit	Qty
02	Android Studio	Latest version	—	1
03	Google Account	—	—	1

3. List of Required Tools & Equipment

S/N	Name	Specification	Unit	Qty
01	Personal Computer or Laptop	Computer	—	1
02	Keyboard and Mouse	Optical mouse	—	1
03	Monitor	Monitor	—	1
04	Internet connection	—	—	1